# Week 9 Homework

**Exercise**

Solve the recurrence relation

$$a_n = \sqrt{\frac{a_{n-2}}{a_{n-1}}}$$

with initial conditions $a_0 = 8$ and $a_1 = 1/(2\sqrt{2})$ by taking the logarithm of both sides and making the substitution $b_n = \log_2 a_n$.

**Exercise**

Solve the recurrence relation

$$a_n = (a_{n-1} \cdot a_{n-2})^2$$

with initial conditions $a_0 = 1$ and $a_1 = \sqrt{2}$ by taking the logarithm of both sides and making the substitution $b_n = \log_2 a_n$.

**Exercise**

Solve the recurrence relation $T(n) = nT^2(n/2)$ with initial condition $T(1) = 6$. (Hint: Let $n = 2^k$ and then make the substitution $a_k = \log T(2^k)$.)

**Exercise**

Convert the following non-homogeneous linear recurrence relations into homogeneous linear recurrence relations by eliminating the non-homogeneous piece.

a. $a_n = 2a_{n-1} + n + 1$

b. $b_n = 3b_{n-1} - 3b_{n-2} + 2^n$

**Exercise**

Find a recurrence relation for $c_n$ the number of ways to construct a $2 \times 2 \times n$ pillar out of $2 \times 2 \times 1$ bricks. What are the initial conditions?

**Exercise**

Find a recurrence relation for $b_n$ the number of ways to construct a $2 \times 2 \times n$ pillar out of $2 \times 1 \times 1$ bricks. What are the initial conditions? (Though very similar to the previous problem, this version is much harder. Hint: The solution is not $b_n = 2b_{n-1} + 5b_{n-2}$, that is only part of the solution.)

**Exercise**

Find a recurrence relation for $d_n$ the number of ways to construct a $2 \times 2 \times n$ pillar out of $2 \times 2 \times 1$ bricks and $2 \times 2 \times 2$ bricks . What are the initial conditions?

**Exercise**

Find the definition of the **complement** $\bar{G}$ of a graph $G$. Give three examples of a graph and it's complement.

**Exercise**

A tree with $n$ vertices is called **graceful** if it's vertices can be labeled with the integers 1, 2 , ..., $n$ such that the absolute values of the difference of the labels of adjacent vertices are all different. Draw and label two different graceful trees on 6 vertices.

**Exercise**

Create a graph to represent all states of the following problem where the vertices are an ordered pair of how much water is each jug (e.g (5,0) means 5 gallons in the 5 gallon jug and 0 gallons in the 3 gallon jug) and the edges represent the change levels with one pour. Problem: "You've got to defuse a bomb by placing exactly 4 gallons of water on a sensor. The problem is, you only have a 5 gallon jug and a 3 gallon jug on hand!"

**Proof**

Prove that if there are two different paths $P_1 : x, v_1, v_2, \cdots, v_k, y$ and $P_2 : x, u_1, u_2, \cdots, u_l, y$ where at least one $v_i \neq u_j$ for some $i$ and $j$between vertices $x$ and $y$ in a simple graph, then there must be a cycle in the graph.

**Proof**

Prove that if a simple graph contains an odd length closed trail $v_0 \to v_1 \to v_2 \to \cdots \to v_k$ then the graph contains an odd length cycle.

**Proof**

Prove $\frac{1}{n+1}\binom{2n}{n} = \frac{2 \cdot 6 \cdot 10 \cdots (4n-2)}{(n+1)!}$. Both of these expressions are closed form solutions to the Catalan number sequence $C_n$.

**Proof**

Let $C_n$ be the Catalan number sequence, prove that $(n+1)C_n = (4n-2)C_{n-1}$.

**Proof**

Prove the sum of the first $n$ triangular numbers is $\frac{n(n+1)(n+2)}{6}$

**Proof**

Prove the that the number of $n$-digit binary numbers with no consecutive 1's is the Fibonacci number $F_{n+2}$, where $F_1 = 1$ and $F_2 = 1$.

**Proof**

Prove the following statement about the Catalan numbers $C_n$ for all all positive integers $n \geq 1$: $C_n \geq 2^{n-1}$ (Hint: You may want to use the last part of Proof .[cproof])

**Proof**

Prove the following statement about the Catalan numbers $C_n$ for all all positive integers $n \geq 1$: $C_n \geq \frac{4^{n-1}}{n^2}$ (Hint: You may want to use the last part of Proof .[cproof]).